

In this problem set, the questions marked with a  $(\star)$  are required for students enrolled in 583 and bonus for students enrolled in 483.

### 1. Differentiation and Transformation of Four-Vectors

Two inertial frames  $S$  and  $S'$  are related by a Lorentz boost with speed  $v$  in the  $+x$ -direction. Units are chosen so that  $c = 1$ . The transformation is

$$t' = \gamma(t - vx), \quad (1)$$

$$x' = \gamma(x - vt), \quad (2)$$

$$y' = y, \quad (3)$$

$$z' = z, \quad (4)$$

where  $\gamma = (1 - v^2)^{-1/2}$ .

A particle moves along the  $x$ -axis in frame  $S$  with worldline

$$x(t) = at^2, \quad y = z = 0,$$

where  $a$  is a constant.

The four-position is  $x^\mu = (t, x, y, z)$ , and the four-velocity is defined as

$$u^\mu = \frac{dx^\mu}{d\tau},$$

where  $\tau$  is the particle's proper time.

- (a) (5 points) Compute  $u^\mu$  in frame  $S$ , expressing your answer in terms of  $t$  and  $a$ . (Hint: first compute  $dx^\mu/dt$ , then use  $d\tau = dt\sqrt{1 - v^2}$ , where  $v = dx/dt$ .)

**Solution:**  $x^\mu = (t, at^2, 0, 0) \implies dx^\mu/dt = (1, 2at, 0, 0)$ . We can use that  $d\tau = dt\sqrt{1 - v^2}$  to get  $dx^\mu/d\tau = dx^\mu/dtdt/d\tau = \gamma(1, 2at, 0, 0)$ .

- (b) (2 points) Show explicitly that

$$u^\mu u_\mu = -1.$$

**Solution:**  $u^\mu u_\mu = \eta_{\mu\nu} t^\mu u^\nu = (1 - v^2)^{-1}(-1 + 4a^2t^2)$ . Identifying  $v = dx/dt = 2at$  gives us  $u^\mu u_\mu = (1 - v^2)^{-1}(-1 + v^2) = -1$ .

- (c) (5 points) Using the Lorentz transformation above, compute the components of  $u'^\mu$  in frame  $S'$ . Verify that

$$u'^\mu u'_\mu = -1,$$

and explain briefly why this result was expected.

**Solution:** The transformation to the primed frame is given by

$$u'^{\mu} = \frac{dx'^{\mu}}{d\tau} = \frac{dx'^{\mu}}{dx_{\nu}} \frac{dx^{\nu}}{d\tau}. \quad (5)$$

where

$$\frac{dx'^{\mu}}{dx^{\nu}} = \begin{pmatrix} \gamma & v\gamma \\ -v\gamma & \gamma \end{pmatrix}. \quad (6)$$

Therefore, using  $v = dx/dt = 2at$ , we have

$$\frac{dx'^{\mu}}{dx_{\nu}} \frac{dx^{\nu}}{d\tau} = \gamma^2 \begin{pmatrix} 1 & 2at \\ -2at & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2at \end{pmatrix} \quad (7)$$

$$= \gamma^2 \begin{pmatrix} 1 + 4a^2t^2 \\ -2at + 2at \end{pmatrix} \quad (8)$$

$$= \frac{1}{1 - 4a^2t^2} \begin{pmatrix} 1 + 4a^2t^2 \\ 0 \end{pmatrix} \quad (9)$$

$$= \begin{pmatrix} -1 \\ 0 \end{pmatrix}. \quad (10)$$

In the above, WLOG, we focus on the first two entries of the four-vector. Thus,  $u'^{\mu} = (-1, 0, 0, 0)$  and  $u'^{\mu}u'_{\mu} = \eta_{\mu\nu}u'^{\mu}u'^{\nu} = -1$ . This result is expected as we have a massive particle, which should be timelike, and the magnitude of the four-velocity is invariant under Lorentz transformations.

(d) (5 points) The four-acceleration is defined as

$$a^{\mu} = \frac{du^{\mu}}{d\tau}.$$

Compute  $a^{\mu}$  for the particle in frame  $S$  and show that

$$u^{\mu}a_{\mu} = 0.$$

**Solution:** Recall that  $u^{\mu} = \gamma(1, 2at, 0, 0)$ . Therefore,

$$\frac{du^{\mu}}{d\tau} = \frac{du^{\mu}}{dt} \frac{dt}{d\tau}. \quad (11)$$

We can calculate each piece of the expression on the RHS, again focusing on the non-zero entries:

$$\frac{du^{\mu}}{dt} = \left( \frac{d\gamma}{dt}, 2at \frac{d\gamma}{dt} + 2a\gamma \right), \quad \frac{d\gamma}{dt} = \gamma \frac{4a^2t}{1 - 4a^2t^2} \quad (12)$$

$$= \gamma \left( \frac{4a^2t}{1 - 4a^2t^2}, \frac{8a^3t^2}{1 - 4a^2t^2} + 2a \right). \quad (13)$$

Identifying that  $dt/d\tau = \gamma$  results in

$$a^\mu = \gamma^2 \left( \frac{4a^2t}{1 - 4a^2t^2}, \frac{8a^3t^2}{1 - 4a^2t^2} + 2a \right). \quad (14)$$

Thus,

$$u^\mu a_\mu = \gamma^3 \left( \frac{-4a^2t}{1 - 4a^2t^2} + \frac{16a^4t^3}{1 - 4a^2t^2} + 4a^2t \right) \quad (15)$$

$$= \frac{\gamma^3}{1 - 4a^2t^2} (-4a^2t + 16a^4t^3 + 4a^2t - 16a^4t^3) \quad (16)$$

$$= 0. \quad (17)$$

- (e) (2 points) Explain why differentiation with respect to proper time preserves the four-vector character of a quantity, whereas differentiation with respect to coordinate time  $t$  generally does not.

**Solution:**  $t$  is a frame-dependent quantity while  $\tau$  is not, i.e.  $\tau$  is a Lorentz scalar while  $t$  is not.

## 2. Tachyons

- (a) (2 points) Argue that a kind of particle that always moves *faster* than the velocity of light would be consistent with Lorentz invariance in the sense that if its speed is greater than light in one frame, it will be greater than light in all frames. (Such a hypothetical particle is called a tachyon).

**Solution:** See handwritten solutions attached.

- (b) (2 points) Show that the tangent vector to the trajectory of a tachyon is spacelike and can be written  $u^\alpha = dx^\alpha/ds$ , where  $s$  is the spacelike interval along the trajectory. Show that  $\mathbf{u} \cdot \mathbf{u} = 1$ .
- (c) (2 points) Evaluate the components of a tachyon's four-velocity  $\mathbf{u}$  in terms of the three velocity  $\vec{V} = d\vec{x}/dt$ .
- (d) (2 points) Define the four-momentum by  $\mathbf{p} = m\mathbf{u}$  and find the relation between energy and momentum for a tachyon.
- (e) (2 points) Show that there is an inertial frame where the energy of any tachyon is negative.
- (f) (2 points) Show that if tachyons interact with normal particles, a normal particle could emit a tachyon with total energy and three-momentum being conserved. This result suggests a world containing tachyons would be unstable and there is no evidence for tachyons in nature.

3. **Relativistic charged particle** Consider the action of a (relativistic) charge particle coupled to a background gauge field,  $A_\mu$ , living in Minkowski space. It is governed by the action

$$S = -m \int d\tau + q \int A_\mu(\mathbf{x}) dx^\mu, \quad (18)$$

where  $\mathbf{A}$  is a 1-form and  $m, q$  are the mass and charge of the particle, respectively.

- (a) (3 points) Suppose that the particle travels along the worldline  $x^\mu(\lambda)$ . Write the action in terms of  $x^\mu$  and  $dx^\mu/d\lambda$ .

**Solution:** First, we can convert all the differentials into  $d\lambda$  by the usual substitution,

$$S = -m \int \frac{d\tau}{d\lambda} d\lambda + q \int A_\mu(\mathbf{x}) \frac{dx^\mu}{d\lambda} d\lambda. \quad (19)$$

We saw in class that  $d\tau/d\lambda = \sqrt{-\eta_{\mu\nu}(dx^\mu/d\lambda)(dx^\nu/d\lambda)}$ , so we have

$$S[x^\mu] = \int d\lambda \left( -m \sqrt{\eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} + q A_\mu(\mathbf{x}) \frac{dx^\mu}{d\lambda} \right). \quad (20)$$

- (b) (5 points) Find the canonical momenta  $p_\mu$  and show that

$$p_\mu = mu_\mu + qA_\mu. \quad (21)$$

In other words, show that the canonical momentum, which is defined by the Lagrangian procedure, is different than the mechanical momentum  $p_\mu = mu_\mu$ .

**Solution:** Let's define  $f$  as the integrand of the action  $S = \int f d\lambda$  so that the canonical momentum is defined as

$$p_\gamma = \frac{\partial f}{\partial(dx^\gamma/d\lambda)}. \quad (22)$$

In this instance,

$$f = -m \sqrt{\eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} + q A_\mu(\mathbf{x}) \frac{dx^\mu}{d\lambda}. \quad (23)$$

Therefore,

$$\begin{aligned}
p_\gamma &= \frac{\partial}{\partial(dx^\gamma/d\lambda)} \left[ -m \sqrt{\eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} + q A_\mu(\mathbf{x}) \frac{dx^\mu}{d\lambda} \right] \\
&= -\frac{m}{2} \frac{1}{\sqrt{-\eta_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda}}} \left( -\eta_{\mu\nu} \delta_\gamma^\mu \frac{dx^\nu}{d\lambda} - \eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \delta_\gamma^\nu \right) + q A_\mu \delta_\gamma^\mu \\
&= \frac{m}{2} \frac{d\lambda}{d\tau} \left( \eta_{\gamma\nu} \frac{dx^\nu}{d\lambda} + \eta_{\gamma\mu} \frac{dx^\mu}{d\lambda} \right) + q A_\gamma \\
&= m \frac{d\lambda}{d\tau} \left( \eta_{\gamma\nu} \frac{dx^\nu}{d\lambda} \right) + q A_\gamma \\
&= m \eta_{\gamma\nu} \frac{dx^\nu}{d\tau} + q A_\gamma \\
&= m u_\gamma + q A_\gamma .
\end{aligned} \tag{24}$$

The desired result follows from relabeling  $\gamma \rightarrow \mu$ .

- (c) (3 points) Complete the Euler-Lagrange equations to show that

$$\frac{dp_\gamma}{d\lambda} = q \frac{\partial A_\mu}{\partial x^\gamma} \frac{dx^\mu}{d\lambda} . \tag{25}$$

**Solution:** To complete the Euler-Lagrange equations  $dp_\gamma/d\lambda = \partial f/\partial x^\gamma$ , we need to find  $\partial f/\partial x^\gamma$ .  $x^\gamma$  only appears in the vector potential so

$$\frac{dp_\gamma}{d\lambda} = \frac{\partial f}{\partial x^\gamma} = \frac{\partial}{\partial x^\gamma} \left[ q A_\mu(\mathbf{x}) \frac{dx^\mu}{d\lambda} \right] = q \frac{\partial A_\mu}{\partial x^\gamma} \frac{dx^\mu}{d\lambda} . \tag{26}$$

- (d) (6 points) (★) Show that your answer in the previous part can be written as

$$m \frac{du_\gamma}{d\tau} = q F_{\gamma\mu} u^\mu , \tag{27}$$

where

$$F_{\gamma\mu} = \partial_\gamma A_\mu - \partial_\mu A_\gamma \tag{28}$$

is the field strength tensor.

**Solution:** Combining our two previous solutions, we have

$$\frac{d}{d\lambda} (m u_\gamma + q A_\gamma) = \frac{\partial A_\mu}{\partial x^\gamma} \frac{dx^\mu}{d\lambda} . \tag{29}$$

Let's change variables from  $\lambda \rightarrow \tau$  by multiplying both sides by  $d\lambda/d\tau$  to get

$$\frac{d}{d\tau}(mu_\gamma + qA_\gamma) = \frac{\partial A_\mu}{\partial x^\gamma} \frac{dx^\mu}{d\tau} = \frac{\partial A_\mu}{\partial x^\gamma} u^\mu. \quad (30)$$

The chain-rule tells us that  $\frac{\partial A_\gamma}{\partial \tau} = \frac{\partial A_\gamma}{\partial x^\mu} \frac{dx^\mu}{d\tau} = \frac{\partial A_\gamma}{\partial x^\mu} u^\mu$  so that

$$m \frac{du_\gamma}{d\tau} + q \frac{\partial A_\gamma}{\partial x^\mu} u^\mu = q \frac{\partial A_\mu}{\partial x^\gamma} u^\mu \implies m \frac{du_\gamma}{d\tau} = q \left( \frac{\partial A_\mu}{\partial x^\gamma} - \frac{\partial A_\gamma}{\partial x^\mu} \right) u^\mu = q F_{\gamma\mu} u^\mu. \quad (31)$$

- (e) (5 points) Show that  $F_{\gamma\mu}$  is antisymmetric and has the correct transformation properties to be a tensor.

**Solution:** The antisymmetric nature of  $F_{\gamma\mu}$  follows from  $F_{\gamma\mu} = \partial_\gamma A_\mu - \partial_\mu A_\gamma$ ; switching  $\gamma$  and  $\mu$  results in an overall minus sign on the right-hand side. We can test the transformation properties by going into the barred frame,

$$\begin{aligned} F_{\bar{\gamma}\bar{\mu}} &= \partial_{\bar{\gamma}} A_{\bar{\mu}} - \partial_{\bar{\mu}} A_{\bar{\gamma}} \\ &= \frac{\partial x^\delta}{\partial x^{\bar{\gamma}}} \frac{\partial (\Lambda_{\bar{\mu}}^\nu A_\nu)}{\partial x^\delta} - \frac{\partial x^\nu}{\partial x^{\bar{\mu}}} \frac{\partial (\Lambda_{\bar{\gamma}}^\delta A_\delta)}{\partial x^\nu} \\ &= \Lambda_{\bar{\gamma}}^\delta \Lambda_{\bar{\mu}}^\nu \frac{\partial A_\nu}{\partial x^\delta} - \Lambda_{\bar{\mu}}^\nu \Lambda_{\bar{\gamma}}^\delta \frac{\partial A_\delta}{\partial x^\nu} \\ &= \Lambda_{\bar{\gamma}}^\delta \Lambda_{\bar{\mu}}^\nu \left( \frac{\partial A_\nu}{\partial x^\delta} - \frac{\partial A_\delta}{\partial x^\nu} \right) \\ &= \Lambda_{\bar{\gamma}}^\delta \Lambda_{\bar{\mu}}^\nu F_{\delta\nu}, \end{aligned} \quad (32)$$

which is the correct transformation law.

- (f) (6 points) (★) By considering the acceleration of slow-moving particles, show that the components of  $F_{\gamma\mu}$  agree with

$$F_{\gamma\mu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}. \quad (33)$$

**Solution:** The spatial components of Eq. 27 are  $mdu_i/d\tau = qF_{i\mu}u^\mu = qF_{i0}u^0 + qF_{ij}u^j$ . The designation “slow-moving” means that  $v \ll 1$  and  $\gamma \simeq 1$ , such that  $u^i \simeq v^i, u^0 \simeq 1$ , and  $d\tau \simeq dt$ . Thus, we have  $mdv^i/dt = qF_{i0} + qF_{ij}v^j$ . This is the force-equation for the particle (Newton's 2nd Law). Recall from electrostatics that stationary particles in an electric field  $E$  feel a force  $F_i = qE_i$

and so we can identify  $F_{i0} = E_i$  and  $F_{0i} = -E_i$ . For a moving particle, there is an additional magnetic force  $q(\vec{v} \times \vec{B})_i = q\epsilon_{ijk}v^jB^k = qF_{ij}v^j$  so we can identify  $F_{ij} = \epsilon_{ijk}B^k$ . Plugging in component-by-component, we end up with Eq. 33.

- (g) (4 points) (★) Write the independent components of Eq. 28 corresponding to  $E_x$  and  $B_x$ . Does this match with what you've learned in E&M of how fields are derived from a vector potential? What is the physical interpretation of the component  $A_0$ ?

**Solution:** For the electric field,  $E_x = F_{01} = \partial_1 A_0 - \partial_0 A_1 = \partial_x A_t - \partial_t A_x$ , which generalizes to  $\vec{E} = \vec{\nabla} A_0 - \partial \vec{A} / \partial t$ . This should look familiar if we identify  $A_0 = \phi$  as the electric potential. For the magnetic field  $B_x = F_{23} = \partial_2 A_3 - \partial_3 A_2 = \partial_y A_z - \partial_z A_y$ , which generalizes to  $\vec{B} = \vec{\nabla} \times \vec{A}$ . This should also look familiar from E&M!

Tachyons Harte 8.23

(a) If a particle has  $v > 1$ , it is still Lorentz invariant iff  $v > 1$  in all frames.

$$\text{suppose } v \equiv dx/dt > 1 \Rightarrow dx > dt \Rightarrow dx^2 > dt^2$$

$$\therefore ds^2 = -dt^2 + dx^2 > 0$$

for Lorentz invariance,  $ds^2 = ds'^2$  so we also have that  $-dt'^2 + dx'^2 > 0$  or  $dx' > dt'$  or  $dx'/dt' > 1$ .

can also try this w/ velocity addition formula:  
which is a result of Lorentz invariance

$$v' = \frac{v-u}{1-uv} \quad \text{and } |v| > 1, |u| < 1 \text{ relative velocity}$$

$$\begin{aligned} \text{suppose } |v'| &= \frac{|v-u|}{|1-uv|} < 1 \Rightarrow |v-u| < |1-uv| \\ &\Rightarrow (v-u)^2 < (1-uv)^2 \\ &\Rightarrow v^2 - 2uv + u^2 < 1 - 2uv + u^2v^2 \\ &\text{or } v^2 + u^2 < 1 + u^2v^2 \\ &v^2 + u^2 - u^2v^2 < 1 \\ &v^2(1-u^2) + u^2 < 1 \\ &v^2(1-u^2) < 1-u^2 \\ &\text{or } v^2 < 1 \quad \nabla \end{aligned}$$

(b) Show that  $u^\alpha = dx^\alpha/d\lambda$  is spacelike and  $u^\alpha u_\alpha = 1$

$$\text{know } v \equiv dx/dt = dx/d\lambda / dt/d\lambda \geq 1$$

$$\text{so } dx/d\lambda \geq dt/d\lambda$$

$$\therefore u^\alpha u_\alpha = -\left(\frac{dt}{d\lambda}\right)^2 + \left(\frac{dx}{d\lambda}\right)^2 \geq 0 \quad \text{or} \quad -dt^2 + dx^2 \geq d\lambda^2$$

Now take  $\lambda = s$  where  $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$

$$\text{then } 1 = g_{\mu\nu} \frac{dx^\mu}{ds} \frac{dx^\nu}{ds} = u^\mu u_\mu$$

(c) evaluate components of tachyon's four-velocity  $u$  in terms of 3-velocity  $\vec{v} \equiv d\vec{x}/dt$

$$x^\mu = (t, \vec{x}), \quad \frac{dx^\mu}{ds} = u^\mu = \left( \frac{dt}{ds}, \frac{d\vec{x}}{ds} \right)$$

$$\begin{aligned} \text{divide by } dt/ds \text{ to get } u^\mu &= \frac{dt}{ds} \left( 1, \frac{d\vec{x}}{ds} \frac{ds}{dt} \right) \\ &= \frac{dt}{ds} (1, \vec{v}) \end{aligned}$$

$$\text{we also have } ds^2 = -dt^2 + d\vec{x}^2 \text{ so } 1 = -\left(\frac{dt}{ds}\right)^2 + \left(\frac{d\vec{x}}{ds}\right)^2$$

$$\text{multiply by } \left(\frac{dt}{ds}\right)^2 \rightarrow \left(\frac{ds}{dt}\right)^2 = -1 + |\vec{v}|^2$$

$$\text{or } \frac{dt}{ds} = (-1 + |\vec{v}|^2)^{-1/2} \Rightarrow \boxed{u^\mu = (-1 + v^2)^{-1/2} (1, \vec{v})}$$

(d) define 4-momentum  $p = m u$ . Find relationship between energy-momentum for tachyon

$$\begin{aligned} p \cdot p &= m^2 \quad p = (E, \vec{p}) \Rightarrow m^2 = -E^2 + |\vec{p}|^2 \\ \text{or } \boxed{E^2 &= |\vec{p}|^2 - m^2} \end{aligned}$$

(e) show that there is an inertial frame where energy of tachyon is negative.

Pick rest frame of tachyon, i.e.  $\vec{p} = 0$ .

$$\text{then } E^2 = -m^2 < 0$$

(f) show if tachyon interacts w/ normal particle, particle can emit tachyon and still preserve E-momentum.

let normal particle have  $\underline{k}_i = (M, \vec{0})$  and it emits a tachyon  $\underline{k} = (E, \vec{p})$ .  $\underline{k}_f = (E_f, \vec{k})$

$$\text{EMC: } \underline{k}_i = \underline{k}_f + \underline{k}$$

$$(\underline{k}_i)^2 = (\underline{k}_f + \underline{k})^2 = -M^2 = \cancel{k_f^2}^{+ \hbar^2} + 2\underline{k}_f \cdot \underline{k} + \underline{k}^2$$

$$\text{or } 2\underline{k}_f \cdot \underline{k} = -\underline{k}^2 = -m^2$$

$$\underline{k}_f \cdot \underline{k} = -E_f E + (\vec{p} \cdot \vec{k}) \quad \text{so} \quad -2E_f E + 2|\vec{p}| |\vec{k}| = -m^2$$

this has solutions ( $-E_f E + |\vec{p}| |\vec{k}|$  can be negative)

b/c  $E$  can be arbitrarily small for fixed  $|\vec{p}|$

and  $\underline{p}$  is spacelike